LAB 02

SERIAL CONNECTION

1. Purpose

- Develop a circuit with some peripherals including RFID reader, real-time IC, graphic OLED display.
- Learn about I2C, SPI serial communication standard protocols, how to address and communicate between single master and multiple slave on the I2C bus.
- Programing the IC DS1307 to get real time data
- Programing the OLED SH1106 graphic display to display data
- Programing the RFID RC522 to detect and read RFID 13.56MHz tags
- Develop an application to simulator door opening and closing based on RFID identify and log opening/closing

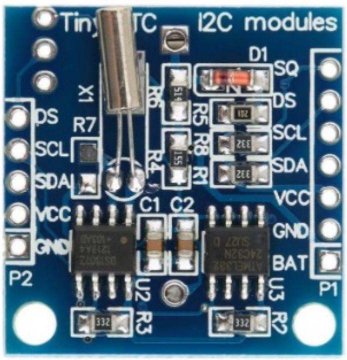
2. Prepare

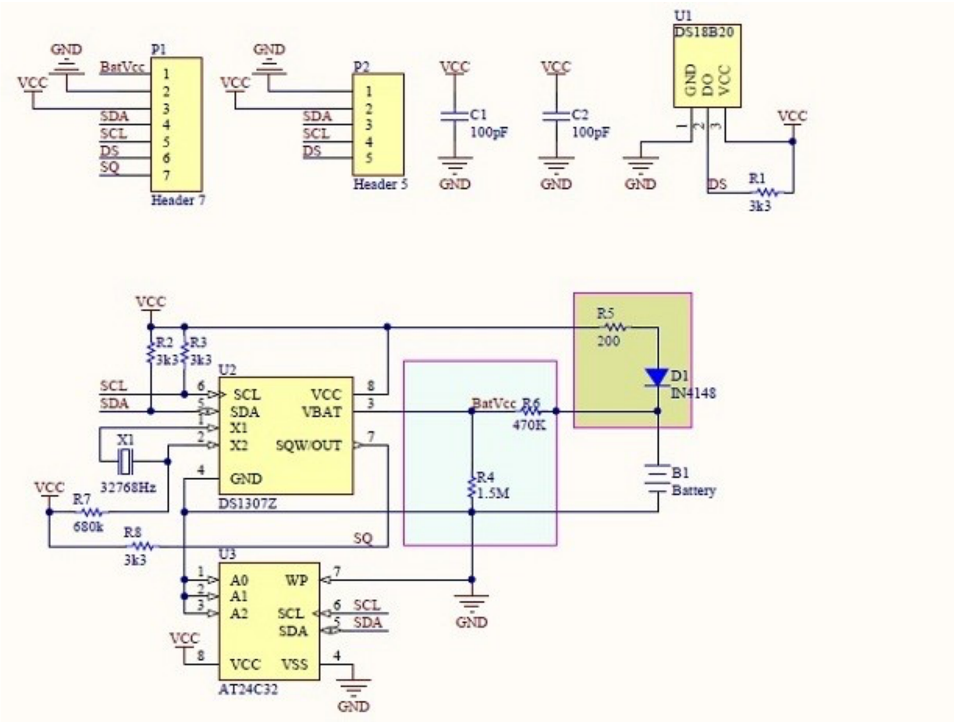
- Hardware: STM32F429I developer, Tiny RTC (DS1307 + AT24C32) module, OLED SH1106 1.3 inches display, RFID RC522 module và RFID 13.56 MHz tags.
- Software: STM32CubeIDE, Hercules.

3. Lab content

3.1. Learn about DS1307 + AT24C32 Tiny RTC module

- This is a small module integrating 3 IC including DS1307, AT24C32 and DS18B20 on the same circuit. This module can perform 3 operations at the same time: providing real-time, storing data and measuring temperature. DS1307 and AT24C32 using I2C protocol.
- Tiny RTC Module and diagram:





Require:

Read DS1307, AT24C32 datasheets and read above diagram answer questions:

- Can these 2 IC work at same time? Why?

Which pins need to be conbected to the Tiny RTC Module

Power pins:……………………………………………..

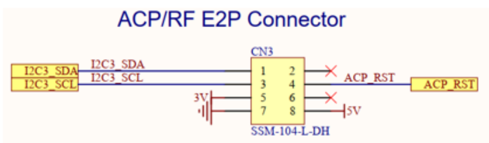
Signal pins:……………………………………………

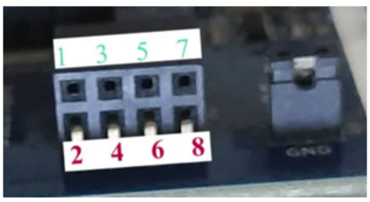
- What are the address of DS1307 and AT24C32?

3.2. Learn about OLED SH1106 module

- This is graphic display module with 128x64 resolution. This module is small and low energy consumption but has high display quality. Besides, SH1106 use I2C bus so just need only 2 signal lines SCL and SDA with high transmission speed.

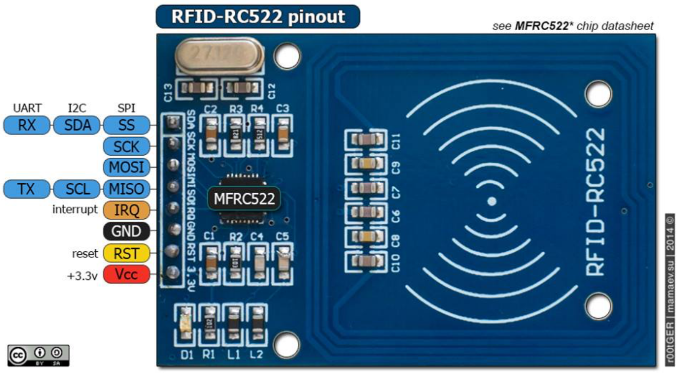
- **NTOE:** There are 2 type of OLED display with opposite order of SCL and SDA





3.3. Learn about RC522 RFID reader module

This module reads/writes data with RFID 13.56MHz tags.



Cho phép ghép nối qua cả 3 chuẩn: UART, I2C, SPI. Ở bài thực hành này, module được hàn cứng các chân cấu hình và chỉ hỗ trợ chuẩn SPI.

Allow to connect via 3 protocols: UART, I2C, SPI. In this laboratory, this module is configured and supports only SPI protocol

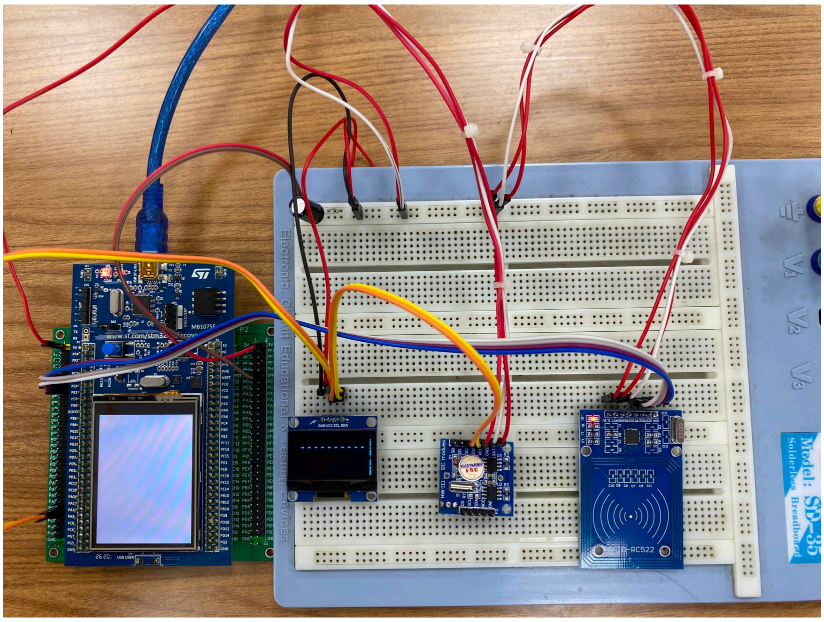
3.4. Diagram design

Design a circuit diagram for the following signal groups:

- Power supply
  - Common ground between modules.
  - SH1106 use 3V power supply from STM32 KIT.
  - TinyRTC:
    - VCC connect with 5V power supply from STM32 KIT.
  - RC522:
    - VCC connect with 3V.
  - 100uF capacitor between 3V and GND.
- Connecting I^2C:
  - SCL, SDA from CN3 or from PA8, PC9 on STM32 KIT.
  - SCL, SDA of Tiny RTC and SH1106.
- Connecting SPI:
  - SS of RC522 connect with PE4 of STM32.
  - SCK of RC522 connect with PE2 of STM32.
  - MISO of RC522 connect with PE5 of STM32.
  - MOSI of RC522 connect with PE6 of STM32.
  - RST of RC522 connect with 3V.

NOTE: *Only supply power to the circuit after the teacher or teacher's assistant has checked to ensure that the circuit is not short-circuited*

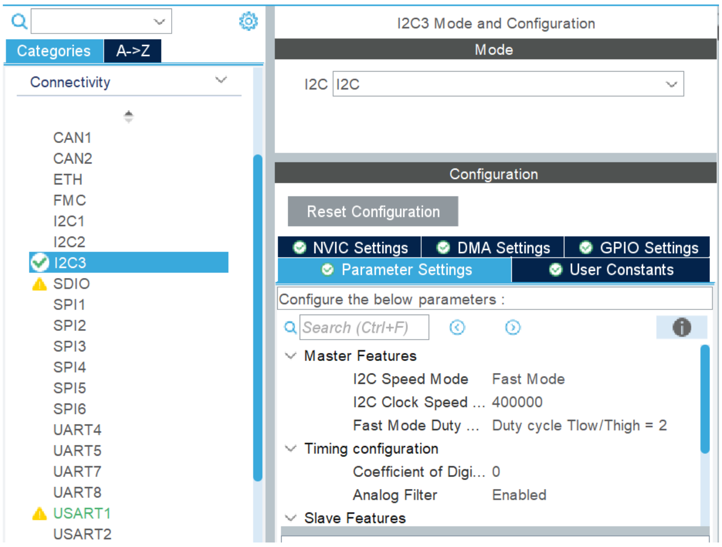
Reference diagram

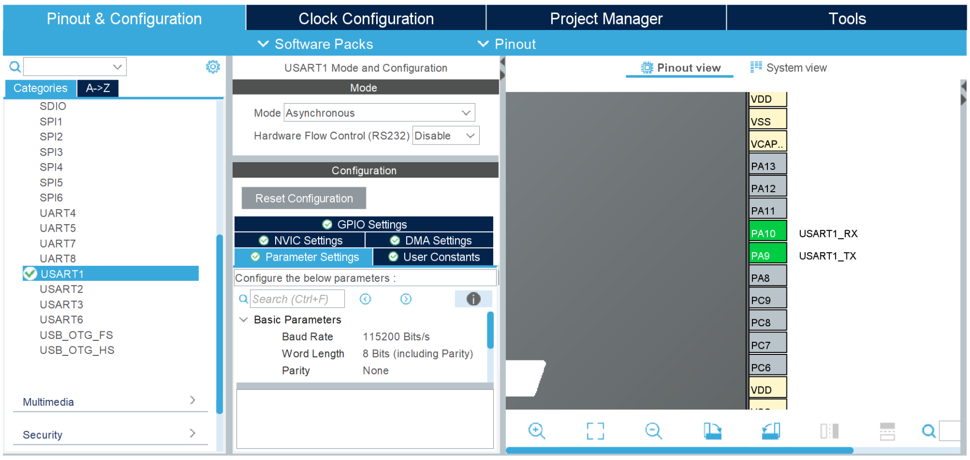


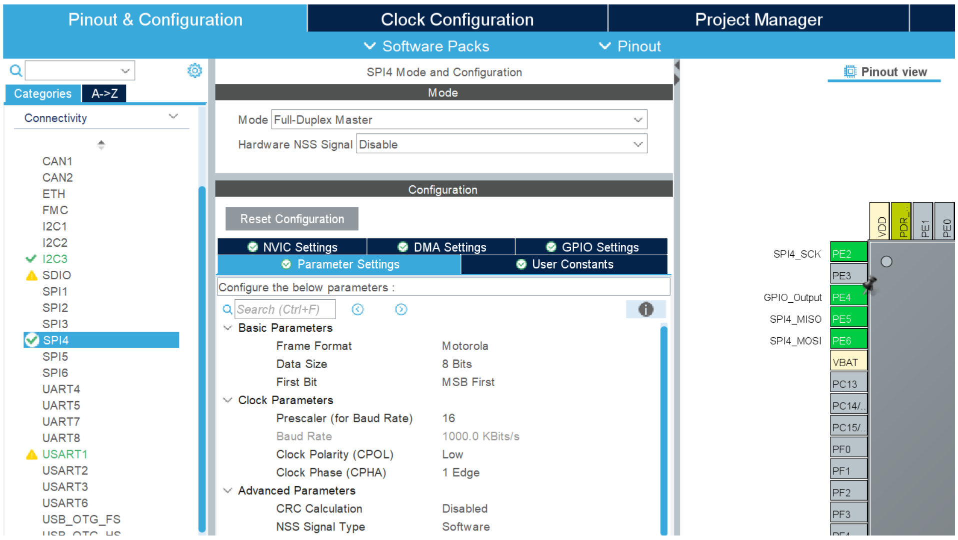
3.5. Configured the connection ports on STM32 Cube IDE

Create new projects and configure

- Turn on RCC and set the CPU clock to 180 MHz.
- Configure PE4 ad GPIO_Output.
- Configure I2C3, USART1, SPI4 peripherals in Connectivity







3.6. Programming connection with DS1307

Add TestDS1307() function below into main.c file. Call TestDS1307() function in main() function.

```
struct Time{
          uint8_t sec;
          uint8_t min;
          uint8_t hour;
          uint8_t weekday;
          uint8_t day;
          uint8_t month;
          uint8_t year;
          };

void TestDS1307()
{
        char buff[30];
        struct Time Set_time, Get_Time;

        Set_time.sec = 10;
        Set_time.min = 30;
        Set_time.hour = 7;
        Set_time.day = 17;
        Set_time.month = 4;
        Set_time.year = 25;
        Set_time.weekday = 0;

        //write time to DS1307
        HAL_I2C_Mem_Write(&hi2c3, 0xD0, 0, 1, (uint8_t *)&Set_time, 7, 1000);
        while (1) {
                //read time back
                HAL_I2C_Mem_Read(&hi2c3, 0xD1, 0, 1, (uint8_t *)&Get_Time, 7, 1000);
```

```
                sprintf(buff,"%02d:%02d:%02d-%02d-%02d/%02d/%02d \n",
                        Get_Time.hour,
                        Get_Time.min,
                        Get_Time.sec,
                        Get_Time.weekday,
                        Get_Time.day,
                        Get_Time.month,
                        Get_Time.year);
                HAL_UART_Transmit(&huart1, buff, strlen(buff), 1000);
                HAL_Delay(500);
        }
}
```

Build and install program into STM32 KIT and observe the results on Hercules Software.

Require: Rewrite TestDS1307() into 2 function with suitable parameters .

- SetTime() : set datetime value for DS1307
- GetTime() : get datetime value from DS1307.

Note: The set time function should only be called once to set the correct time. The DS1307 will then maintain the time itself as long as VBAT is powered.

3.7. Programming connection with SH1106

- Add sh1106.* và fonts.* files into project.
- Add code segment below into main() function at appropriate place (After MX_I2C3_Init calling).

```
char buf[100];
uint8_t X = 0, Y = 0;
SH1106_Init ();
sprintf (buf, "%s", "RC522 RFID");
SH1106_GotoXY (12,10); // goto 10, 10
SH1106_Puts(buf, &Font_11x18, 1);
SH1106_UpdateScreen(); // update screen
HAL_Delay(1000);
```

Observe the results show on SH1106 display.

Require: Combine with the above section, write program continuous get datetime value from DS 1307 and display datetime value into SH1106

3.8. Programming connection with RC522

- Add **tm_stm32f4_mfrc522.c** và **tm_stm32f4_mfrc522.h** files into project.
- Add source code into main() function

```
TM_MFRC522_Init();

/* USER CODE END 2 */

/* Infinite loop */
/* USER CODE BEGIN WHILE */
while (1)
{
    uint8_t CardID[5];
    HAL_Delay(100);

    if (TM_MFRC522_Check(CardID) == MI_OK) {
        sprintf (buf, "%s", "Card found");
        SH1106_GotoXY (12,10); // goto 10, 10
        SH1106_Puts(buf, &Font_11x18, 1);
        SH1106_UpdateScreen(); // update screen
    }
    else{
        sprintf (buf, "%s", "----------");
        SH1106_GotoXY (12,10); // goto 10, 10
        SH1106_Puts(buf, &Font_11x18, 1);
        SH1106_UpdateScreen(); // update screen
    }
  /* USER CODE END WHILE */

  /* USER CODE BEGIN 3 */
}
/* USER CODE END 3 */
```

- Build and install program into STM32 KIT and observe the results on SH1106 display when swiping the RFID tags through the reader.

Require: Displays the tag value (5 bytes) that was read into the SH1106 display

3.9. Self - Made Excercies

Combine all above section and develop a firmware on STM32 with function:

- Every time an RFID tag is swiped through the reader, LED3 turns on until the card is removed (card detected notification).

- If the tag code matches a pre-set tag code list, LED4 turns on (simulating the card opening operation) and the SH1106 screen displays “Welcome”. At the same time, a log record including: door opening time and tag code. If the tag code does not match, the SH1106 screen displays “Rejected”.

- Log is saved in RAM, including maximum 100 records.

- Users can communicate with the circuit to set the unlock card code, view the log content by sending commands to the circuit via UART and Hercules on PC. It is necessary to design the protocol, command set, and data exchange method to do this function.